

Question	Answer	Marks
1(a)(i)	$v = r\omega$	C1
	$= 0.85 \times 140$	A1
	$= 120 \text{ m s}^{-1}$	
1(a)(ii)	$a = r\omega^2$ or $a = v^2 / r$	C1
	$a = 0.85 \times 140^2$ or $120^2 / 0.85$	A1
	$= 1.7 \times 10^4 \text{ m s}^{-2}$	
1(b)(i)	direction of (induced) e.m.f.	M1
	is such as to (produce effects that) oppose the <u>change</u> that caused it	A1
1(b)(ii)	$T = 2\pi / \omega$	A1
	$= 2\pi / 140 = 0.045 \text{ s} = 45 \text{ ms}$	
1(b)(iii)	$\Phi = BA$	C1
	$= 0.18 \times \pi \times 0.85^2$	C1
	$= 0.41 \text{ Wb}$	A1
1(b)(iv)	$E = \Phi / t$	C1
	$= 0.41 / 0.045$	A1
	$= 9.1 \text{ V}$	
1(b)(v)	force (on spoke) must be anticlockwise, so current is from A to X (by Fleming's left hand rule), so X is at the higher potential	B1

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